Executive Summary

This thesis examines whether networks built from U.S. equity returns can help forecast realized market volatility over the next 30 trading days. Standard volatility approaches usually rely on either past volatility or market-implied expectations. Historical models look backward, while implied-volatility measures such as the VIX look forward through option-market prices. Both are useful and difficult to beat, but they may miss information contained in the changing relationships among individual stocks.

The main idea is that the stock market is not just a list of separate companies. Stocks move together, and the strength of those relationships changes over time. In calm periods, diversification works better because firms are less synchronized. During stress, stocks often move together more strongly, diversification weakens, and the market behaves more like one connected system. If these network changes appear before volatility fully rises, they may provide early signals of future market risk.

This is the gap addressed in the thesis. Existing research shows that volatility is persistent, that the VIX is a strong benchmark, and that connectedness rises during stress. However, much of the financial-network literature remains descriptive. It often shows that networks look different during crises, but not whether networks can forecast future volatility before it happens. This thesis therefore asks whether dated stock-return networks contain useful out-of-sample predictive information for future market volatility.

The topic also reflects my interest in applying machine learning to a finance problem where the structure of the data matters. I wanted to apply neural-network methods to a setting where relationships between observations are central. Equity markets are a natural case because stocks are connected through sector exposure, investor behavior, common shocks, and market-wide stress. Over time, I began to think of the market less as a flat table of returns and more as a living network whose structure expands, contracts, and reorganizes.

To test this idea, the thesis implements a modular empirical pipeline. The analysis constructs graphs for a stable universe of 500 U.S. stocks. Firms are the nodes, and links are formed from rolling return correlations. These graphs are used to train a Graph Neural Network that produces one forecast per date for realized volatility over the next 30 trading days, measured from daily S&P 500 returns.

Forecasts are evaluated in a strict chronological out-of-sample design, meaning that the model is tested on later data it did not see during training. The main benchmark is the VIX, while the thesis also compares the model with a HAR-type historical-volatility benchmark and

a placebo-graph ablation. The VIX is used only as a benchmark and is not included as an input to the graph model. This makes the test demanding: the graph model must forecast using equity-return structure rather than option-market information.

The results show that the graph-based model contains clear out-of-sample predictive information about future realized market volatility. Relative to the VIX, the model performs better on log-scale forecast errors in the full test sample and in several restricted-sample robustness exercises. This means that information from stock-return networks can be turned into useful forward-looking volatility forecasts.

The results are encouraging, but they are not overstated. The graph model does not beat the VIX on every measure. In the full test sample, the VIX performs better under one variance-based loss measure, and the evidence that the graph model adds stable information beyond the VIX is borderline rather than definitive. The diagnostics show a clear pattern: the graph model performs especially well in low- and medium-volatility environments, while the VIX remains more informative in the highest-volatility regime.

The thesis therefore does not argue that the graph model replaces the VIX. The VIX remains essential because it reflects option-market expectations, hedging demand, and tail-risk pricing. The graph model provides a different type of signal. It is built from the internal structure of the equity market rather than from option prices. Its value is strongest as a complementary risk-monitoring tool.

The practical implication is that graph-based forecasts are best used within a broader risk-monitoring framework, rather than as substitutes for existing volatility indicators. Historical models capture persistence in past volatility, the VIX captures option-market expectations and tail-risk pricing, and the graph model captures the evolving connectedness of individual stocks. Although this thesis tests only U.S. equities, the same logic could be extended to crypto markets, where liquidity shocks, leverage cycles, regulation, and sentiment can quickly increase cross-asset co-movement.

Overall, this thesis provides evidence that networks built from U.S. equity returns are useful for forecasting future market volatility. Its main contribution is not to show universal dominance over implied-volatility benchmarks, but to show that stock-return networks can generate useful forward-looking forecasts within a coherent out-of-sample framework. More broadly, the thesis shows that market risk is not only visible in past volatility or option prices, but also in the way individual stocks become connected over time. In short, the evolving structure of the market itself contains information that can improve how future volatility is monitored and forecasted.